



Introduction to Data Mining and Knowledge Discovery

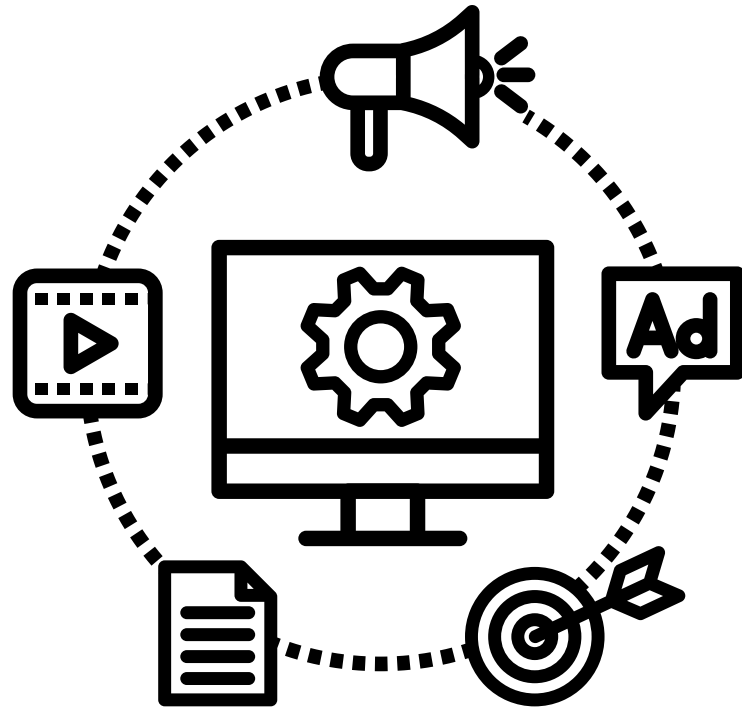
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01 What is Data Mining?

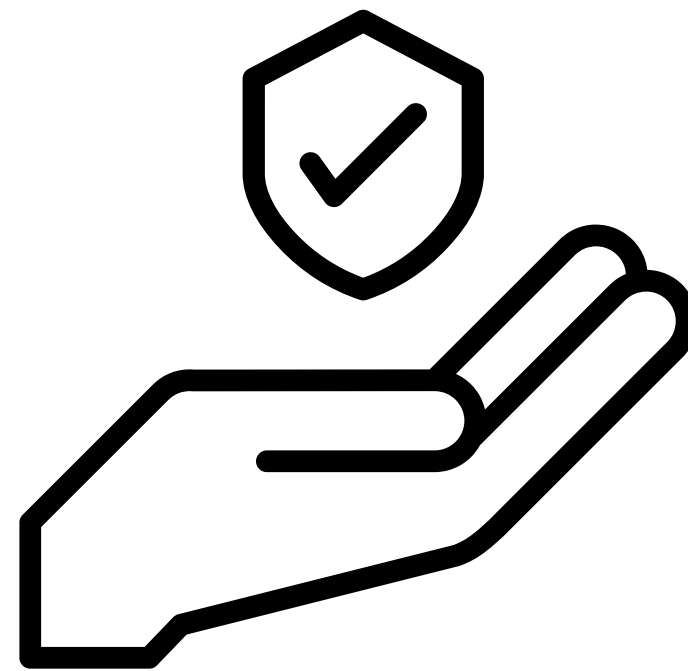
Data mining is the process of sorting through large data sets to identify patterns and relationships that can help solve business problems through data analysis.

Uses of Data Mining



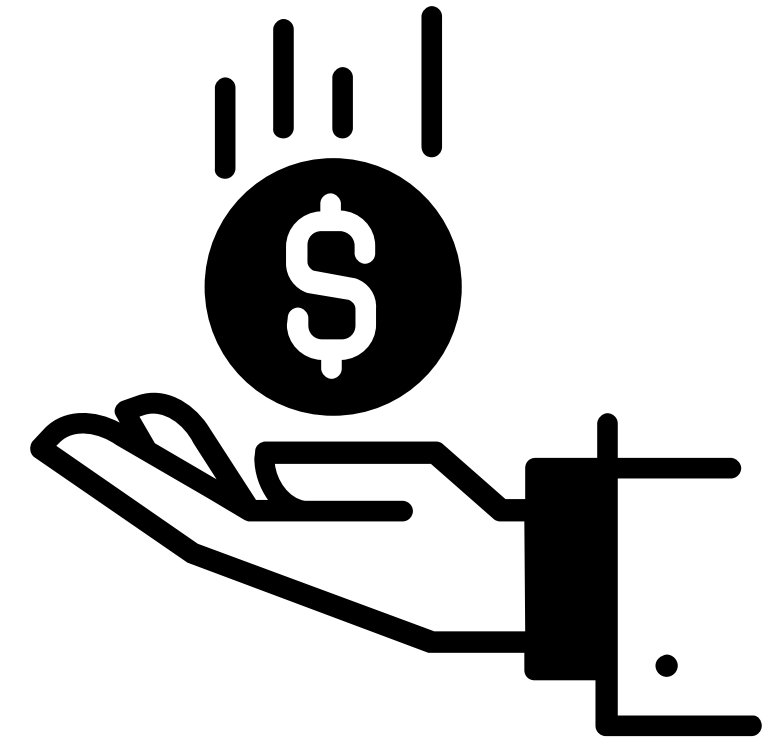
Telecommunications and Media

Telecommunication companies can use customer data to predict customer behavior and offer targeted ads and campaigns



Insurance

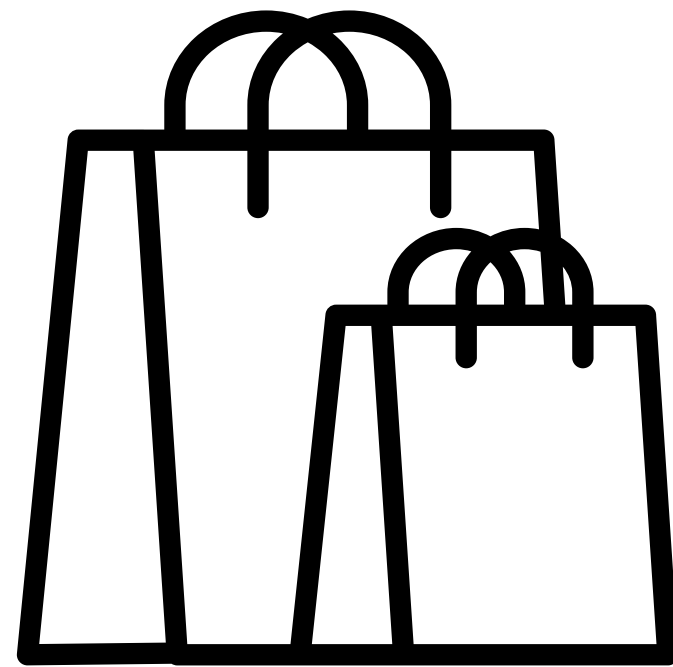
Insurance companies can create complex models for detecting fraudulent claims, risk management, and customer compliance



Finance

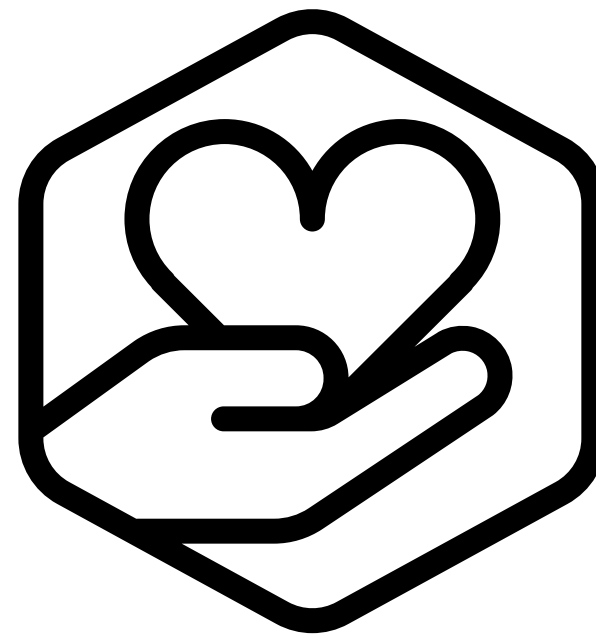
Financial companies make use of data mining for better market risk evaluation and maximize stock market returns

Uses of Data Mining



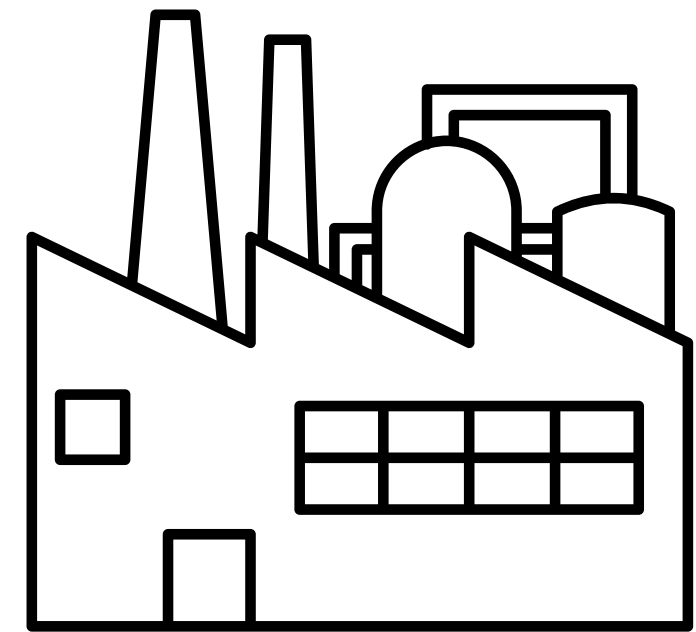
Retail

Supermarkets use joint purchasing patterns to identify product associations and decide how to place them in aisles and shelves



Healthcare

Data mining helps doctors create more accurate diagnoses using patient medical history, physical examination results, medications, and treatment patterns.



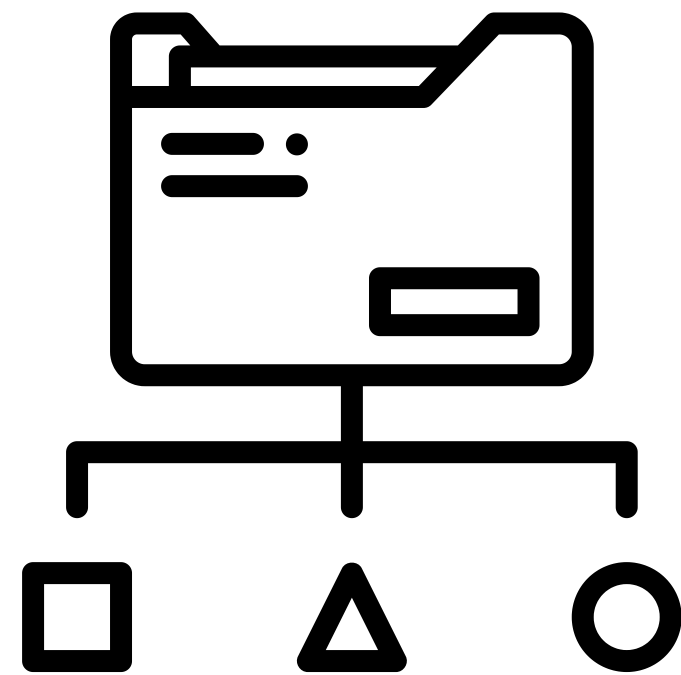
Manufacturing

Implement just-in-time fulfillment by predicting when new supplies should be ordered or when equipment is likely to fail.

Common Data Mining Techniques 02

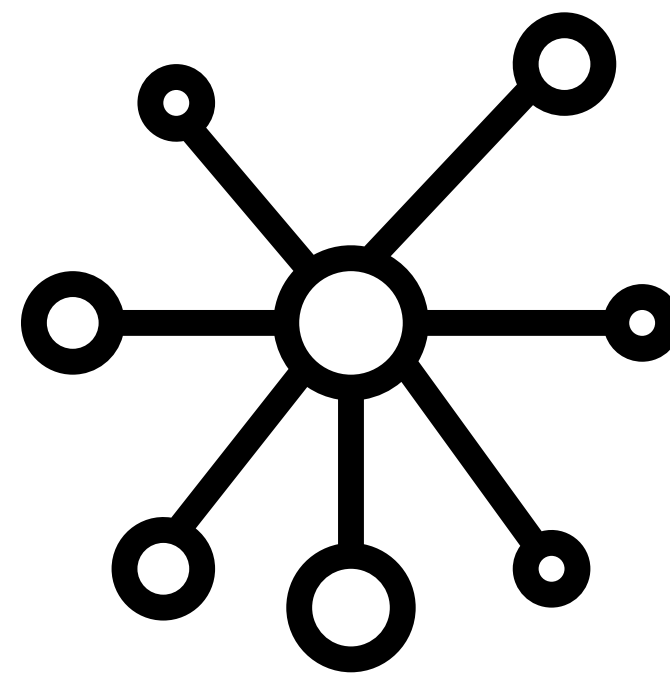
- Classification
- Clustering
- Prediction
- Regression
- Association Rules

Types of Data Mining



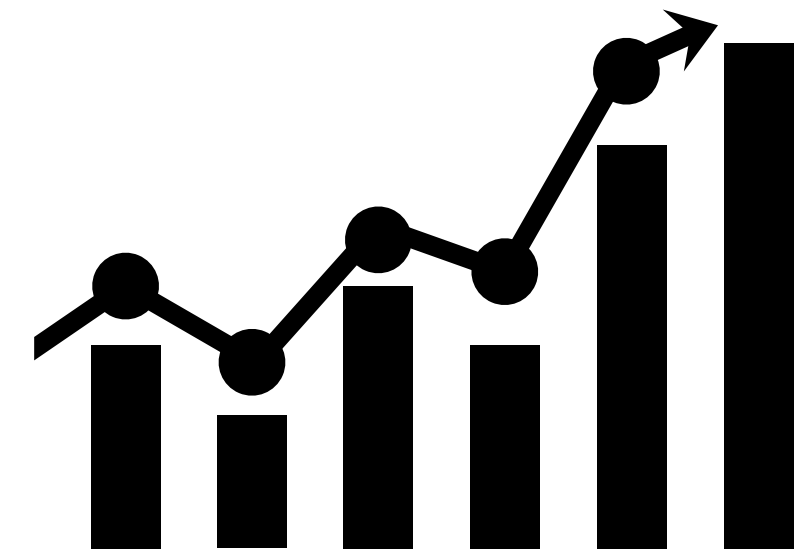
Classification

Classification is a process in which data points from large data sets are assigned to categories based on how they're being used.



Clustering

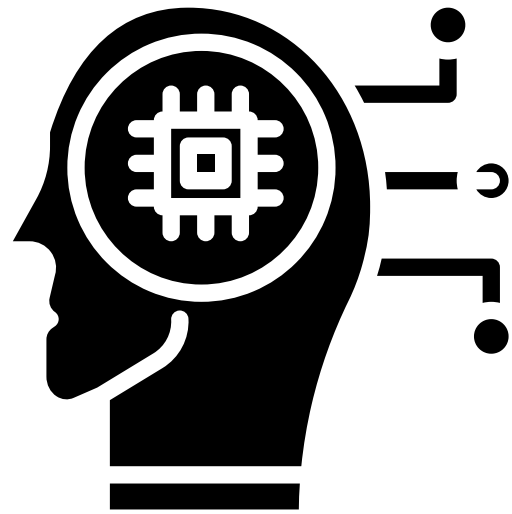
Clustering refers to the process of grouping a series of different data points based on their characteristics.



Regression

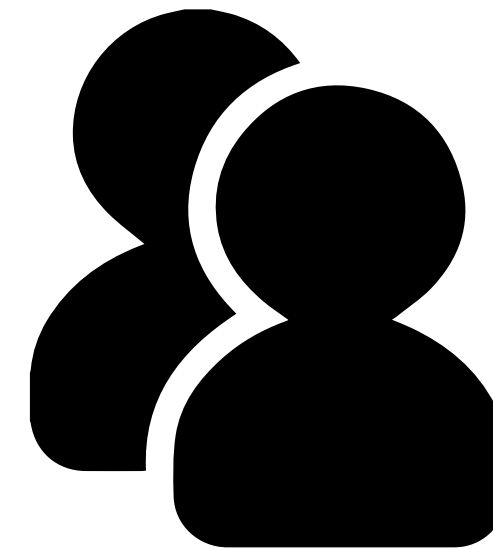
Regression is used to identify and analyze the relationship between variables because of the presence of the other factor.

Types of Data Mining



Prediction

Prediction examine data sets to find patterns and trends, then calculate the probabilities of a future outcome.



Association Rules

Association rules are used to find correlations, or associations, between points in a data set.



Thank You



Sources:

https://www.sas.com/en_ph/insights/analytics/data-mining.html#:~:text=Data%20mining%20is%20the%20process,relationships%2C%20reduce%20risks%20and%20more.
<https://www.sap.com/insights/what-is-data-mining.html>
<https://observationbaltimore.com/blog/data-mining-why-is-it-important/>
<https://light-it.net/blog/reasons-why-companies-are-obsessed-with-data-mining/>
<https://www.loginworks.com/blogs/217-data-mining-and-its-importance/>
<https://www.iberdrola.com/innovation/data-mining-definition-examples-and-applications>
<https://bootcamp.pe.gatech.edu/blog/10-key-data-mining-techniques-and-how-businesses-use-them/#:~:text=There%20are%20numerous%20crucial%20data,%2C%20neural%20networks%2C%20and%20prediction.>
<https://www.javatpoint.com/data-mining-techniques>
<https://www.netsuite.com/portal/resource/articles/data-warehouse/data-mining.shtml>